Studies for Identification of the Inrush Based on Improved Correlation Algorithm

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***Abstract*—A new scheme based on an improved correlation algorithm to distinguish between inrush and internal fault in the transformer is proposed. In this scheme, the current over one cycle of each phase is reorganized as a new measurand. Fault currents can be distinguished from the inrush current if the improved waveform correlation coefficient between the first half-cycle and the latter-half-cycle of the above measurand is over a threshold. Test results show that all types of internal faults, even those accompanied by the magnetizing inrush, can be correctly identified from the inrush with the proposed algorithm. Besides high security, this scheme has satisfactory sensitivity. In addition, this algorithm can be easily implemented in the real-time process.**

***Index Terms*—Improved waveform correlation algorithm, inrush, internal fault, transformer differential protection.**

I. Introduction

DIFFERENTIAL protective relay has been used as the primary protection of most power transformers for many years. Inrush can be generated when a transformer is switched on the transmission line or an external line fault is cleared. It may result in mal-operation of differential protection if blocking scheme is unavailable. Therefore, to distinguish between inrush and fault current is the key to improve the security of the differential protection.

Three types of schemes are currently in use for this purpose. Some schemes make use of the information obtained from the differential currents of the transformer, such as the method based on the scheme of second harmonic restraint [1]. Some make use of the information obtained from the variation of the transformer terminal voltages, such as the method based on the voltage restraint principle [2]. Other alternative schemes make use of the information obtained from both the currents and the voltages of the transformer, such as the method based on the flux characteristic principle and the method based on the equivalent circuit equation of the transformer model [3], [4]. However, the most widely used methods in practice are still those which are based on the principle of the second harmonic restraint. The main drawback of this method is that the harmonics occurring during the internal fault can cause differential relay either not to operate or to operate with a long time delay [5]. Furthermore, according to the current schemes using only a single measurand, no matter whether it is current or voltage, and cross-blocking principle, any internal fault with energization cannot be cleared quickly until the inrush fades out, and thus, the differential relay is unblocked. Some versions such as adaptive threshold on energization, cross-phase blocking, and adaptive cross-phase blocking are utilized to remedy blocking scheme based on magnitude percentage of second harmonics [6]. The above schemes still deserve to be improved although they are effective in the field applications. For example, the cross-blocking principle based scheme of adaptive threshold on energization will still delay the operation of the differential relay when a faulted transformer is energized, and the cross-phase blocking-based scheme cannot operate quickly when a transformer with light winding faults is energized. Some novel technique such as the one that uses a complex second harmonic ratio (both magnitude and angle) and a 2-D comparator has been proposed recently, showing that the identification of inrush has been an active research field.

Recently, a novel blocking scheme called “waveform symmetry,” or “waveform correlation,” has been proposed [6], which compares the symmetry between the first-half-cycle and the latter-half-cycle waveform in a total one-cycle evaluating time window span. This scheme deserves to be studied further since it considers the shape, magnitude, and regularity of the current waveform over all. With this scheme, the performance of differential protection is expected to be enhanced further, e.g., it is possible to implement the pole-separated restraint for the relay without decreasing the security. However, there are still some drawbacks to be overcome, such as the selection of compared signals and the expression of waveform correlation coefficient. It is shown in this paper that the security and sensitivity of this scheme has been improved with the proposed algorithm. Encouraging results from the ATP software-based simulations and tests on a physical power system in the laboratory environment are obtained and presented in the paper.

II. Basic Principle of Waveform Correlation Scheme

The waveform correlation scheme (WCS) to distinguish between inrush and fault current is presented with Fig. 1 in this section.

This scheme will take effect as soon as a pickup signal is issued, and one-cycle samplings have been available in the data window. This sampled one-cycle data ($0 \sim 20$ ms for 50-Hz power frequency) is extended into a two-cycle sampling ($0 \sim 40$ ms), as shown in Fig. 1(a). From a certain point (t_1), a new one-cycle data is taken with the large box consisting of two small boxes in Fig. 1(a). Let the first-half-cycle signal in the former small box be $x(t)$, whereas let the signal in

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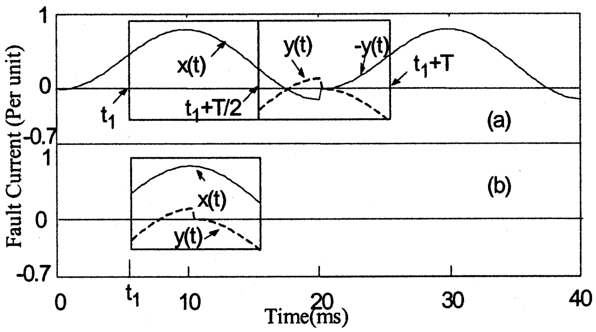

Fig. 1. Internal fault current extended periodically.

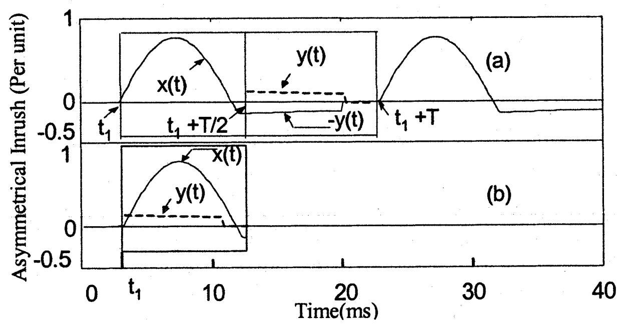

Fig. 2. Inrush with single peak value extended periodically.

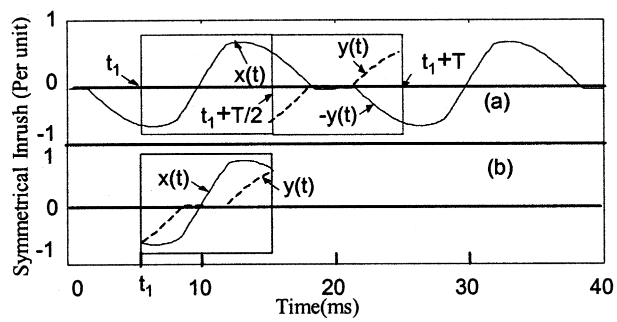

Fig. 3. Inrush with double peak values extended periodically.

the latter one with the same length as the former one be $-y(t)$. The correlation between $x(t)$ and $y(t)$, shown as the dotted-line waveform in the latter small box, was evaluated in [6], and the inrush can be identified from the internal fault current with a low value of the correlation coefficient(CC). In order to illustratively evaluate the degree of correlation between $x(t)$ and $y(t)$, $y(t)$ can be translated forward half-cycle, and then, $y(t)$ and $x(t)$ are located on the same box, as seen in Fig. 1(b). As mentioned previously, the determination of t_1 and the constitution of CC are the most important to the performance of WCS. The scheme in [6] adopts maximum area algorithm to determine t_1.

Actually, the signal shown in Fig. 1 is an excursive internal fault current of which the original value of the damped dc component is equal to the amplitude of the periodical component, and the sampling rate is N times the fundamental. A half cycle integral window, shown as the small box including $x(t)$ in Fig. 1(a), is shifted from 0 to 20 ms, Consequently, total N area values are calculated by means of the absolute value sum of the current sampling in this time window, which is shown as

$$S(k) = \sum_{j=k}^{k+N/2-1} |i(\mathrm{mod}(j))| \; k = 0, 1, 2, \ldots N-1 \tag{1}$$

where

$$\mathrm{mod}(j) = \begin{cases} j & j < N \\ j - N & j > N-1 \end{cases}. \tag{2}$$

The term "$\mathrm{mod}\,(j)$" is the subscript of i and is applicable for (5). Let the origin of the time window corresponding to the maximum among these N area values be t_1, from which a one-cycle signal is sampled, and the correlativity between $x(t)$ and $y(t)$ is evaluated. In Fig. 1(b), the correlativity is rather well for this internal fault. In contrast, the corresponding situations of two types of typical inrushes are shown in Figs. 2 and 3, respectively. It can be seen in Figs. 2 and 3 that $x(t)$ is quite different from $y(t)$ for both types of the inrushes, that is, the relativity between them is weaker than that of the fault current.

A revised correlation coefficient, called waveform coefficient here, is derived from the above analysis:

$$J = \frac{\mathrm{Cov}(X,Y)}{\sigma^2(X)}. \tag{3}$$

In contrast, a normal correlation coefficient is given by

$$J = \frac{\mathrm{Cov}(X,Y)}{\sigma(X)\sigma(Y)} \tag{4}$$

where Cov (X,Y) is the covariance of $x(t)$, and $y(t)$, $\sigma(X)$, and $\sigma(Y)$ denote mean-square deviation of $x(t)$ and $y(t)$. It is shown that the difference between waveform coefficient J and normal correlation coefficient ρ_{XY} is that $\sigma(X)\sigma(Y)$ in ρ_{XY} is substituted by $\sigma^2(X)$. With such substitution, the difference of J between inrush and fault current becomes more distinct as $\sigma(Y)$ is much smaller than $\sigma(X)$ when evaluating the inrush in Fig. 2, and the $\sigma(Y)$ is approximately equal to $\sigma(X)$ when analyzing the current in Fig. 1. Criterion 1 is derived according to (1) and (3). The value J is calculated as soon as the differential protection is initiated, and the protection will operate if J is over a threshold; otherwise, the protection will be blocked.

Criterion 1 appears to be a promising amelioration for the security of differential protection. However, it is rather hard to be set in certain cases, as those illustrated by the simulations below.

The simulations are based on ATP software [8]. Some important parameters of the simulated transformer are shown in Table I. Besides, the CT model with 1:1 ratio is introduced. The modeling method of CT and the CT's parameters are all from [9] in which the hysteresis model of CT is simulated as a main loop and a set of minor loop trajectory. See [9] for more details. In the simulations, the transformer is energized from the high-voltage side since it is a step-up transformer, and there is a generator circuit breaker on the low-voltage side. A second-harmonic based restraint scheme and dwell-time approach are introduced as a coordinate system to highlight the effectiveness of WCS.

TABLE I
PARAMETERS OF THE SIMULATED TRANSFORMER

Transformer Type	3- phase, 2-winding transformer bank
Connection style	HV side WYE-connected, LV side DELTA-connected, the neutral terminal on high voltage side grounded (YNd5)
Saturation flux density	Bs=1.15Bm
Rated apparent power	750MVA
Short-circuit power	4600MVA
Voltage ratio	420kV/27kV
Core model	Type-96

First of all, an extreme test, of which the residual fluxes are $B_{\rm ra} = 0.9B_m$, $B_{\rm rb} = B_{\rm rc} = -0.9B_m$, is carried out, and the inrushes sampled from the primary and secondary sides of the CTs are shown in Figs. 4 and 5, respectively, where the inception angle of phase A voltage source, whose type is sine, is 30°.

The analytical results of Figs. 4 and 5 are seen in Table II. For succinctness of tabling, I_2/I_1 is the percentage of the second harmonic to the fundamental, and the noncurrent range of the dwell-time approach is $\delta_{\rm dw}$, which is denoted by the angle corresponding to noncurrent in one cycle (360°). In addition, J is assumed as the waveform coefficient.

It should be mentioned that $\delta_{\rm dw}$ of the inrush on secondary may fade due to the reverse charge of CT. To regain $\delta_{\rm dw}$ when the dwell-time approach is implemented in the micro-processor based protection, a differentiation algorithm and a low-current threshold are necessary. Here, the low-current threshold, which is the proportion between sampled value and the maximum of the absolute value in one cycle, is set to 0.008. With the above threshold, $\delta_{\rm dw}$ is only equal to 1.8° in one cycle (360°) for a standard sinusoidal wave.

As for the other denotations of Table II, "side" indicated the inrushes to be analyzed are sampled whether from the primary or the secondary of CT's, "P" means primary and "S" denotes "Secondary." Because of the WYN5 connection of the transformer, the analyzed signal should be the difference between two phases, as shown in the column indexed by "Phase."

As seen, the largest J is 0.80 under such a test, the ratio of second harmonic of inrushes on secondary are all smaller than 15%, and the minimum is 6%. Correspondingly, the minimum $\delta_{\rm dw}$ is 18°, which is much smaller than 60° (one sixth of a cycle in each cycle). Thus, second harmonic restraint and dwell-time approach-based schemes will mal-operate under this test, and the WCS-based scheme has enough security with threshold above 0.80. Further extreme tests are also made, and the indexes of the above schemes are calculated and shown in Table II.

Based on Table II, under some conditions, the second harmonic-based scheme of 15% restraint ratio and dwell-time approaches of 60° $\delta_{\rm dw}$ may mal-operate, even using the cross-blocking principle. By the way, the necessary high sampling rate and the difficulty of setting appropriate low-current threshold are two negative factors to apply the dwell-time approach to the microprocessor-based protection. As seen in Table II, the maximum of J is 0.8. Thus, the security of the WCS-based scheme using the pole-separated tripping principle can be guaranteed if the threshold of J is larger than 0.8.

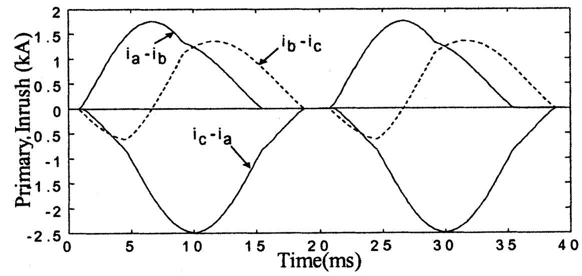

Fig. 4. Three phase magnetizing inrushes on the primary of CT.

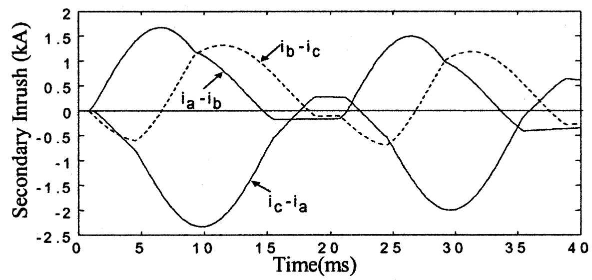

Fig. 5. Three phase magnetizing inrushes on the secondary of CT.

TABLE II
INDEXES OF SECOND HARMONIC RESTRAINT, DWELL TIME APPROACH, AND WCS-BASED SCHEMES UNDER EXTREME TESTS

Residual Flux	Inception angle	Side	Phase	I_2/I_1 (%)	δ_{dw} (°)	J
B_{ra}= 0.9B_m, B_{rb}=-0.9 B_m, B_{rc} =-0.9 B_m	θ_A=0°	P	A-B	12	54	0.65
			B-C	29	30	0.59
			C-A	12	54	0.65
		S	A-B	12	12	0.73
			B-C	29	30	0.58
			C-A	12	36	0.59
	θ_A=30°	P	A-B	12	93	0.54
			B-C	24	33	0.39
			C-A	6.9	42	0.71
		S	A-B	11	93	0.65
			B-C	9	36	0.21
			C-A	6	18	0.80
B_{ra}= 0.9B_m, B_{rb}= 0.0 B_m, B_{rc} =-0.9 B_m	θ_A=0°	P	A-B	8.3	78	0.79
			B-C	35	84	-0.01
			C-A	12	54	0.65
		S	A-B	8.3	69	0.80
			B-C	35	87	-0.09
			C-A	12	36	0.59
	θ_A=30°	P	A-B	12	102	0.56
			B-C	12	102	0.56
			C-A	7	42	0.71
		S	A-B	12	18	0.66
			B-C	12	78	0.48
			C-A	7	18	0.73

Nevertheless, a long time delay is unavoidable using such a high setting value when the internal fault current is distorted seriously. To illustrate this situation, an example of dynamic test is given below.

Fig. 6 shows the configuration and parameters of the dynamic simulation model installed at the laboratory of the Huazhong University of Science and Technology.

Consider a 100% "B"-earth internal fault on the primary of a sound transformer, whose data sampled by the fault recorder

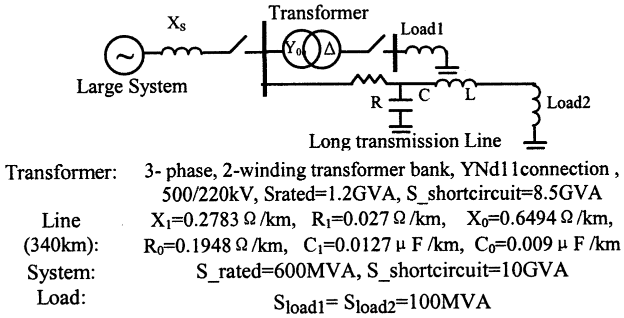

Fig. 6. Dynamic test model and parameters.

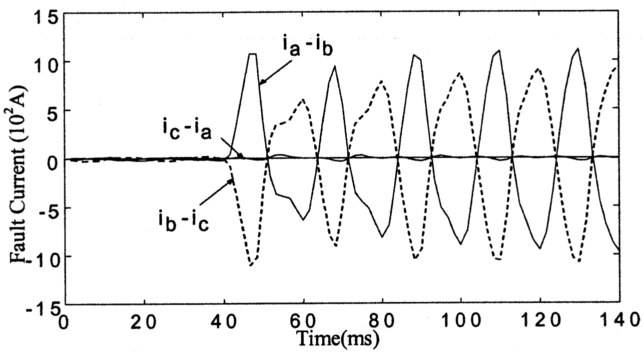

Fig. 7. T “B’”earth fault on primary, with a long line connected.

TABLE III
CHANGES OF WAVEFORM COEFFICIENTS AND RATIOS OF 2ND HARMONIC TO THE FUNDAMENTAL OF THE FAULT CURRENTS

Criteria \ Post-fault time	90	91.7	93.3	95
$(I_2/I_1)_{a\text{-}b}$	15%	15%	14%	13%
$(I_2/I_1)_{b\text{-}c}$	14%	14%	14%	14%
$J_{a\text{-}b}$	0.76	0.76	0.79	0.82
$J_{b\text{-}c}$	0.75	0.75	0.79	0.81

installed on the above system. In this case, the sampling rate is 600 Hz at 50 Hz fundamental frequency, where the phase currents are scaled up currents from the secondary winding of the delta-connected current transformers on the primary terminals of the power transformer.

As shown in Fig. 7, the differential currents exist in both phase A and B due to the delta connection on CTs secondary and the waveforms of currents deform due to the CT saturation. The changes of waveform coefficients and ratios of the second harmonic to the fundamental of the fault currents are illustrated in Table III.

It can be seen from Table III that the protection will not trip until the fault continues for 95 ms using criterion 1 with threshold 0.8. By the way, for the scheme of second harmonic restraint with 15% restraint ratio, the nonoperation time for such fault is 93.3 ms, i.e., the scheme of second harmonic restraint with 15% restraint ratio and the criterion 1 with 0.8 waveform coefficient threshold are approximately equivalent for such a heavy fault.

It is no doubt that the relay will operate more quickly if the threshold of J is decreased. However, the security may be decreased. Therefore, criterion 1 deserves to be improved in sensitivity without decreasing security. Some discussion on how to determine t_1 and improve the constitution of waveform coefficient is given in what follows.

III. DESIGN AND TEST OF THE IMPROVED WAVEFORM CORRELATION PRINCIPLE

A. Improvement of Compared Signal Selection

As mentioned previously, the original maximum area algorithm to separate a one-cycle signal into two parts differing from each other as much as possible is not applicable to distinguish real inrush and internal fault. To solve this problem, the formula of area given by (1) is revised in this paper as

$$S(k) = \sum_{j=k}^{k+N/2} |i(\mathrm{mod}(j))| \; k = 0, 1, 2, \ldots N-1 \qquad (5)$$

where the integral window is prolonged from $N/2$ to $N/2+1$. The ability of the improved maximum area algorithm to separate a signal into two parts with most differences is weakened. However, this exactly meets the demands for distinguishing between the real inrush and the deformed internal fault current. In contrast to (1), (5) is insensitive to the even inrush; therefore, the waveform coefficient has no obvious increase. The improved algorithm also is insensitive to the distorted internal fault current, i.e., J does not apparently decrease, whereas J decreases when using (1) to determine t_1, which results in that the above internal fault will not be distinguished from inrush for a long time of post-fault. Thus, (5) can effectively improve the sensitivity of identifying the real inrush from the fault current.

Equations (3) and (5) are combined as criterion 2. The waveform coefficients of three phase inrushes in Fig. 5 are $J_{a-b} = 0.652$, $J_{b-c} = 0.211$, $J_{c-a} = 0.804$, that is, the security of criterion 2 for inrush is approximately equivalent to that of criterion 1. In contrast to criterion 1, by use of the fault in Fig. 7, the performance of criterion 2 is evaluated with a 0.8 waveform coefficient threshold. Here, the nonoperation time of criterion 2 is 40 ms since J_{a-b} increases to 0.83 at $t_{post-fault} = 40$ ms. At this time, the ratio of I_2/I_1 using Fourier algorithm is 31%, and the differential protection with 15% second harmonic restraint ratio cannot operate. It is clear that criterion 2 can shorten the nonoperation time by about 2.7 cycles.

B. Sensitivity Analysis of Criterion 2

Table IV shows the nonoperation times of criteria 1 and 2 using and pole-separated trip mode and 0.8 threshold the scheme of second harmonic restraint using cross-blocking mode and 15% threshold for all types of typical internal faults, where the fault data came from the previoulsly-mentioned fault recorder. Based on the analyses in Section II and the above thresholds, the securities of criteria 1 and 2 is higher than that of second harmonic restraint, even though the pole-separated tripping is applied to criteria 1 and 2, whereas the cross-blocking mode is used for second harmonics restraint. In the Table IV, N.O.T denotes nonoperation time, and NOP means nonoperation, i.e., the protection still cannot operate, even though the fault has been present for five cycles. For convenience, the faults in Table IV are numbered to indicate the fault conditions. For instance, cases

TABLE IV
NONOPERATION TIMES IN MILLISECONDS FOR 0.8 THRESHOLD OF WAVEFORM COEFFICIENT OF CRITERION 1, CRITERION2, AND 15% OF SECOND HARMONIC RESTRAINT

Case	Fault type and conditions	N.O.T of criterion 1 (ms)	N.O.T Of criterion 2 (ms)	N.O.T of I_2/I_1 (ms)
1	BG_PL	95	40	81.6
2	BG_PUL	68.3	21.6	60
3	BG_UPL	46.6	26.6	58.3
4	BG_UPUL	18.3	16.6	18.3
5	BC_PL	48.3	51.6	93.3
6	BC_PUL	43.3	40	63.3
7	BC_UPL	83.3	43.3	78.3
8	BC_UPUL	40	31.6	60
9	ab_PL	38.3	20	58.3
10	ab_PUL	28.3	20	20
11	ab_UPL	40*	51.6	73
12	ab_UPUL	36.6	20	58.3
13	BW4_PL	30	30	31.6
14	BW4_PUL	31.6	31.6	31.6
15	BW4_UPL	18.3	20	20
16	BW4_UPUL	25	25	25
17	BW2_PUL	21.6	18.3	20
18	BW2_UPUL	23.3	23.3	26.6
19	I_BG_UPL	81.6	55	NOP
20	I_BG_UPUL	60	20	NOP
21	I_BC_UPL	50	36.6	NOP
22	I_BC_UPUL	40	38.3	NOP
23	I_ab_UPL	25*	38.3	NOP
24	I_ab_UPUL	40	33.3	NOP
25	I_BW4_UPL	25	25	NOP
26	I_BW4_UPUL	25	25	NOP
27	I_BW2_UPUL	23.3	23.3	NOP

1 to 4 are "B"-earth faults on primary, whose operating conditions are that the transformer is loaded with a long line (PL), loaded without a long line (PUL), unloaded with a long line (UPL), and unloaded without a long line (UPUL), respectively. The illustrations of the operating conditions for other faults are similar to the above. Cases 4 through 8 are "B"-"C" phase faults on primary, cases 9 through 12 are "A"-"B" phase faults on secondary, cases 13 through 16 are "B"-phase 4.35% winding faults on primary, and cases 17 and 18 are "B"-phase 2.18% winding faults on primary. Moreover, cases 19 through 27 are energizations with internal faults.

It should be indicated that the above test results for transformer fault and the ones below for transformer energizations are all determined with a variety of random points on wave closures.

The following conclusions can be drawn from Table IV:

1) Both criteria 1 and 2 can always trip quite quickly for the light winding faults, regardless of whether they are accompanied by energization. The longest time delay is 31.6 ms. However, the protection with second harmonic ratio restraint cannot trip for cases 19 through 27, even though the internal faults have present for five cycles.
2) Criterion 2 can operate faster than criterion 1, especially for heavy faults or energization with heavy faults.
3) In some cases, criterion 1 can operate more quickly than criterion 2 when the harmonics contained in the fault current do not influence the evenness of the current. However, only four cases appear in the above 27 cases, and two of them trip faster by only one or two sampling intervals than ones using criterion 2.
4) For the very heavy faults (cases 5,19), both criteria operate with a long time delay (at least 50 ms). Of course, the second harmonic restraint-based scheme also cannot operate at this moment. Therefore, It is still necessary to improve criterion 2 further.

C. *Improvement of the Waveform Coefficient J*

It can been observed from (3) that the waveform coefficient only uses the information of $\sigma(X)$, i.e., the mean square error of "large area" in Fig. 2, and the information of $\sigma(Y)$ (small area) is ignored. As in the analysis in part II, the $\sigma(Y)$ of inrush is quite small. For the fault current containing certain damped dc component, $\sigma(Y)$ may be larger than $\sigma(X)$ according to the investigations for various internal fault waveforms. Therefore, an improved waveform coefficient is given as

$$J = \frac{\mathrm{Cov}(X,Y)}{\sigma^2(X)} \times \frac{\sigma^2(Y)}{\sigma^2(X)}. \tag{6}$$

Equations (5) and (6) form criterion 3. With this criterion, the waveform coefficient of inrush will decrease due to the introduction of $\sigma(Y)$. For criterion 3, the waveform coefficients of the inrushes in Fig. 5 are evaluated as $J_{a-b} = 0.523$, $J_{b-c} = 0.181$, $J_{c-a} = 0.661$. As seen, the maximum J has decreased to 0.661. In fact, the inrushes in Fig. 5 only indicate the very extreme case. In this case, the maximum ratio of I_2/I_1 among three phases only is given by 11%, whereas the accepted ratio in the industrial applications is 15 to 20% in China. Consequently, the waveform coefficient J can be decreased so that this algorithm can be contrasted with the scheme of second harmonic restraint. The threshold of criterion 3 can be decreased to $0.66 * 11/15$, i.e., 0.48. In order to compare the criteria 1–3 with the same thresholds induced from Fig. 5, criterion 3 with threshold 0.48 is named criterion 4. Furthermore, the general setting rule is recommended with (7)

$$J = \frac{0.66*11}{(100*K_{21})} \tag{7}$$

where K_{21} is the general setting of the ratio between 15 and 30% for industrial applications. The security and sensitivity evaluations of all the above four criteria are described in what follows.

Fig. 8 illustrates the nonoperation times in milliseconds for the faults in Table IV, where the X-axis is denoted as fault number, and the Y-axis is denoted as the nonoperation time.

The nonoperation time series for criterion is denoted by marker "+," and the ones for criterion 2–4 by "*," "o," "△," respectively. As seen, criterion 4 obviously operates more quickly than the other three criteria. For most faults, the relay can trip within one cycle after a fault occurs. In these 27 cases, the quickest operation time is 13.3 ms, and the longest one is 21.6 ms, and any winding fault can be cleared with one cycle even though the short circuit is present before the transformer is energized. It can be concluded form Fig. 8 that criterion 2 has nearly the same sensitivity as criterion 3. However, criterion 3 has better security than criterion 2 when they are applied to identify the inrushes. The maximum waveform coefficients of all types of typical inrushes recorded by the same recorder were investigated. Among them, ten values are given in Table V. Of which, cases 1-4 are such conditions that

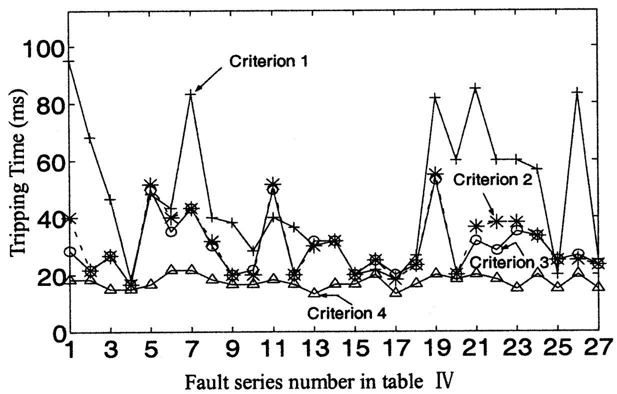

Fig. 8. Nonoperation times of criterion 1 through criterion 4.

the transformer is energized without a long transmission line, and cases 5-9 are that the transformer is energized with a long line. The energization induced by the clearance of external fault is given by case 10. The maximum of waveform coefficients and I_2/I_1 of all cases during the five cycles after inrushes occur are indicated in Table V, where $J_{\max}$ is the maximum waveform coefficient among three phases within five cycles of post-closure, and $K_{\max}$ is the maximum I_2/I_1 among three phases at the same interval.

It can been seen from Table V that the maximum $J_{\max}$ among ten cases are 0.55 or 0.54 and 0.18 for criteria 1-3, respectively, and the minimum $K_{\max}$ among the same analytical samples is 0.30.

For convenient contrast, a reliability coefficient is defined as

$$k_{\text{rel}} = \frac{J_{\text{theomax}}}{\text{Max}(J_{\text{realmax}})}. \tag{8}$$

where J_{theomax} is the theoretical maximum waveform coefficient induced by the inrushes in Fig. 5, and $\text{Max}(J_{\text{realmax}})$ is the real maximum waveform coefficient among the values in Table V. It is evident that K_{rel} is 1.45, 1.48, 3.67, and 2.67, respectively. The security of criteria 3 and 4 is far larger than that of criteria 1 and 2, and criterion 4 is more sensitive than criterion 3 because of a smaller threshold of J. In contrast, the reliability of second harmonic restraint using cross-blocking principle can be evaluated with

$$k_{\text{rel}} = \frac{\text{Min}(K_{\text{realmax}})}{K_{\text{theomax}}}. \tag{9}$$

The definition of K_{realmax} and K_{theomax} is similar to J. In this case, K_{realmax} and K_{theomax} are equal to 0.30 and 0.11, respectively. Thus, the K_{rel} is 2.72, which is approximately equivalent to the criterion 4.

In order to validate the proposed algorithms more thoroughly, some further work should be advanced. For one thing, more types of ATP simulation-based transformers, such as 3-leg or 5-leg Core Type ones, should be considered. Besides, more dynamic tests in the laboratories other than the one of HUST should be done. As a prototype, the company for which the author is working has implemented this algorithm in its transformer protection product. Such a protection device will be put into commission as soon as it passes the dynamic simulation tests at the laboratory of Shangdong University, Shangdong, China.

TABLE V
MAXIMUMS OF WAVEFORM COEFFICIENTS OF REAL MAGNETIZING INRUSHES DERIVED BY CRITERIA 1–3

Case	Inrush Condition	$J_{\max}$ of criterion 1	$J_{\max}$ of criterion 2	$J_{\max}$ of criterion 3	$K_{\max}$ of I_2/I_1
1	Inrush_UL1	0.5	0.46	0.11	0.48
2	Inrush_UL2	0.5	0.5	0.14	0.37
3	Inrush_UL3	0.5	0.46	0.11	0.42
4	Inrush_UL4	0.55	0.51	0.16	0.49
5	Inrush_L1	0.55	0.54	0.18	0.54
6	Inrush_L2	0.41	0.41	0.08	0.36
7	Inrush_L3	0.44	0.44	0.10	0.38
8	Inrush_L4	0.51	0.5	0.15	0.30
9	Inrush_L5	0.48	0.48	0.13	0.45
10	External Fault	0.55	0.46	0.11	0.51

IV. CONCLUSIONS

An improved scheme (criterion 4) used for the transformer inrush identification of differential protection is developed in the paper. The proposed scheme is based on improved correlation analysis. Simulation and dynamic test results show that this algorithm is effective in distinguishing inrushes from different types of internal faults of a transformer with or without transmission lines and loads. The scheme has high security when the threshold is determined according to the boundary inrush condition. On the other hand, the sensitivity of the scheme also is higher than that of other common schemes with the same security. Furthermore, the computation simplicity assures that this scheme can be implemented in the real-time applications with the present microprocessor hardware.

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